Lab Sheet 2: DML Commands – INSERT, UPDATE and DELETE

SQL Answers and Expected Outputs (MySQL Workbench)

Note: These are example queries and illustrative outputs. Run Lab Sheet 1 first and ensure the listed tables/columns exist. Some statements assume Student has Email and Phone, and Employee has Email. Use a fresh practice database because UPDATE and DELETE change data.

1. Insert one record into the Student table.

SQL Query:

```
INSERT INTO Student (Student_ID, Name, Age, Course, City)
VALUES (1, 'Aman', 20, 'BCA', 'Meerut');
```

Expected Output:

```
Query OK, 1 row affected

SELECT * FROM Student;
+------------+------+-----+--------+--------+
| Student_ID | Name | Age | Course | City   |
+------------+------+-----+--------+--------+
| 1          | Aman | 20  | BCA    | Meerut |
+------------+------+-----+--------+--------+
```

2. Insert five student records using separate INSERT statements.

SQL Query:

```
INSERT INTO Student VALUES (2, 'Riya', 21, 'BCA', 'Delhi');
INSERT INTO Student VALUES (3, 'Karan', 20, 'BSc', 'Noida');
INSERT INTO Student VALUES (4, 'Neha', 22, 'BCom', 'Meerut');
INSERT INTO Student VALUES (5, 'Rahul', 21, 'BCA', 'Delhi');
INSERT INTO Student VALUES (6, 'Simran', 20, 'BSc', 'Ghaziabad');
```

Expected Output:

```
Each statement: Query OK, 1 row affected
Total inserted: 5 rows
```

3. Insert multiple records into the Student table using a single statement.

SQL Query:

```
INSERT INTO Student (Student_ID, Name, Age, Course, City) VALUES
(7, 'Arjun', 19, 'BCA', 'Delhi'),
(8, 'Pooja', 20, 'BSc', 'Meerut'),
(9, 'Vikas', 21, 'BCom', 'Noida');
```

Expected Output:

```
Query OK, 3 rows affected
```

4. Insert records into the Employee table.

SQL Query:

```
INSERT INTO Employee (Emp_ID, Emp_Name, Department, Salary, Email) VALUES
(101, 'Amit', 'IT', 45000, 'amit@example.com'),
(102, 'Priya', 'HR', 40000, 'priya@example.com');
```

Expected Output:

```
Query OK, 2 rows affected
(Assumes Employee has Emp_ID, Emp_Name, Department, Salary and Email columns.)
```

5. Insert records into the Department table.

SQL Query:

```
INSERT INTO Department (Dept_ID, Dept_Name, Location) VALUES
(1, 'IT', 'Delhi'),
(2, 'HR', 'Meerut'),
(3, 'Sales', 'Noida');
```

Expected Output:

```
Query OK, 3 rows affected
```

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6. Insert records into the Product table.

SQL Query:

```
INSERT INTO Product (Product_ID, Product_Name, Price, Quantity) VALUES
(1, 'Keyboard', 800, 15),
(2, 'Mouse', 450, 20),
(3, 'Monitor', 7500, 0);
```

Expected Output:

```
Query OK, 3 rows affected
```

7. Insert records into the Customer table.

SQL Query:

```
INSERT INTO Customer (Customer_ID, Customer_Name, Email, Phone) VALUES
(1, 'Ankit Sharma', 'ankit@example.com', '9876543210'),
(2, 'Meera Singh', 'meera@example.com', '9876501234');
```

Expected Output:

```
Query OK, 2 rows affected
```

8. Insert records into the Course table.

SQL Query:

```
INSERT INTO Course (Course_ID, Course_Name, Duration, Fees) VALUES
(1, 'BCA', '3 years', 60000),
(2, 'BSc Data Science', '3 years', 80000);
```

Expected Output:

```
Query OK, 2 rows affected
```

9. Insert a student record with values for all columns.

SQL Query:

```
INSERT INTO Student (Student_ID, Name, Age, Course, City, Email, Phone)
VALUES (10, 'Ishita', 20, 'BCA', 'Delhi', 'ishita@example.com', '9876512345');
```

Expected Output:

```
Query OK, 1 row affected
(Assumes Student currently has Email and Phone columns.)
```

10. Insert a student record by specifying only selected columns.

SQL Query:

```
INSERT INTO Student (Student_ID, Name, Course)
VALUES (11, 'Dev', 'BCA');
```

Expected Output:

```
Query OK, 1 row affected
Unspecified columns receive their DEFAULT value or NULL if allowed.
```

11. Update the city of a particular student.

SQL Query:

```
UPDATE Student SET City = 'Delhi' WHERE Student_ID = 1;
SELECT Student_ID, Name, City FROM Student WHERE Student_ID = 1;
```

Expected Output:

```
Query OK, 1 row affected
+------------+------+-------+
| Student_ID | Name | City  |
+------------+------+-------+
| 1          | Aman | Delhi |
+------------+------+-------+
```

12. Update the salary of a particular employee.

SQL Query:

```
UPDATE Employee SET Salary = 50000 WHERE Emp_ID = 101;
SELECT Emp_ID, Emp_Name, Salary FROM Employee WHERE Emp_ID = 101;
```

Expected Output:

```
Query OK, 1 row affected
+--------+----------+--------+
| Emp_ID | Emp_Name | Salary |
+--------+----------+--------+
| 101    | Amit     | 50000  |
+--------+----------+--------+
```

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13. Increase the salary of all employees by 10%.

SQL Query:

```
UPDATE Employee SET Salary = Salary * 1.10;
SELECT Emp_ID, Emp_Name, Salary FROM Employee;
```

Expected Output:

```
Query OK, 2 rows affected (example data)
Amit: 55000.00; Priya: 44000.00
```

14. Increase the price of all products by 5%.

SQL Query:

```
UPDATE Product SET Price = Price * 1.05;
SELECT Product_ID, Product_Name, Price FROM Product;
```

Expected Output:

```
Query OK, 3 rows affected (example data)
Keyboard: 840.00; Mouse: 472.50; Monitor: 7875.00
```

15. Change the course of a particular student.

SQL Query:

```
UPDATE Student SET Course = 'BSc Data Science' WHERE Student_ID = 1;
SELECT Student_ID, Name, Course FROM Student WHERE Student_ID = 1;
```

Expected Output:

```
Query OK, 1 row affected
Student 1 (Aman): BSc Data Science
```

16. Update the department of an employee.

SQL Query:

```
UPDATE Employee SET Department = 'Finance' WHERE Emp_ID = 102;
SELECT Emp_ID, Emp_Name, Department FROM Employee WHERE Emp_ID = 102;
```

Expected Output:

```
Query OK, 1 row affected
Employee 102 (Priya): Finance
```

17. Update the city of all students from Meerut to Delhi.

SQL Query:

```
UPDATE Student SET City = 'Delhi' WHERE City = 'Meerut';
SELECT Student_ID, Name, City FROM Student WHERE City = 'Delhi';
```

Expected Output:

```
Query OK, rows affected depends on matching records
All students previously in Meerut now show Delhi.
```

18. Update the fees of a particular course.

SQL Query:

```
UPDATE Course SET Fees = 65000 WHERE Course_ID = 1;
SELECT Course_ID, Course_Name, Fees FROM Course WHERE Course_ID = 1;
```

Expected Output:

```
Query OK, 1 row affected
+-----------+-------------+-------+
| Course_ID | Course_Name | Fees  |
+-----------+-------------+-------+
| 1         | BCA         | 65000 |
+-----------+-------------+-------+
```

19. Update multiple columns of a student simultaneously.

SQL Query:

```
UPDATE Student
SET Age = 21, Course = 'BCA', City = 'Noida'
WHERE Student_ID = 1;
SELECT Student_ID, Name, Age, Course, City FROM Student WHERE Student_ID = 1;
```

Expected Output:

```
Query OK, 1 row affected
Student 1: Aman, 21, BCA, Noida
```

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20. Delete a particular student record using Student_ID.

SQL Query:

```
DELETE FROM Student WHERE Student_ID = 11;
SELECT * FROM Student WHERE Student_ID = 11;
```

Expected Output:

```
Query OK, 1 row affected
Empty set (0 rows returned)
```

21. Delete all students belonging to a particular city.

SQL Query:

```
DELETE FROM Student WHERE City = 'Ghaziabad';
```

Expected Output:

```
Query OK, number of rows deleted depends on matching students
```

22. Delete employees whose salary is below a specified amount.

SQL Query:

```
DELETE FROM Employee WHERE Salary < 30000;
```

Expected Output:

```
Query OK, number of rows deleted depends on the employee data
```

23. Delete products whose quantity is zero.

SQL Query:

```
DELETE FROM Product WHERE Quantity = 0;
SELECT * FROM Product WHERE Quantity = 0;
```

Expected Output:

```
Query OK, 1 row affected (with sample data)
Empty set (0 rows returned)
```

24. Delete records satisfying multiple conditions using AND/OR.

SQL Query:

```
DELETE FROM Student
WHERE (City = 'Delhi' AND Age < 20)
   OR Course = 'BCom';
```

Expected Output:

```
Query OK, number of rows deleted depends on matching records
Tip: Run the same WHERE clause with SELECT * first to verify which rows will be deleted.
```

25. Perform a complete sequence of INSERT, UPDATE and DELETE operations on a Student table and display the table after each operation.

SQL Query:

```
INSERT INTO Student (Student_ID, Name, Age, Course, City)
VALUES (50, 'Test Student', 20, 'BCA', 'Meerut');
SELECT * FROM Student WHERE Student_ID = 50;

UPDATE Student SET City = 'Delhi' WHERE Student_ID = 50;
SELECT * FROM Student WHERE Student_ID = 50;

DELETE FROM Student WHERE Student_ID = 50;
SELECT * FROM Student WHERE Student_ID = 50;
```

Expected Output:

```
After INSERT: row 50 appears with City = Meerut.
After UPDATE: row 50 appears with City = Delhi.
After DELETE: Empty set (0 rows returned).
```

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